Day 2 — Why Earth has Seasons

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will explain that Earth's ~23.5° axial tilt — not its distance from the Sun — is what causes the seasons, and demonstrate this with a torch-and-globe model.

Materials

- A small globe (or a ball with N and S marked) — one per group ideally, one as demo minimum
- A strong torch / table lamp (the "Sun")
- A protractor (to verify the tilt angle)
- A blackout / curtained room if possible, OR a hallway
- Student notebooks

Vocabulary introduced today

- **Axial tilt** (obliquity of the ecliptic) — Earth's rotation axis is tilted ~23.5° from a line perpendicular to its orbital plane. This tilt does not change over a year — Earth keeps its tilt pointed in the same direction in space as it goes around the Sun.

(No new Sanskrit term today — yesterday's *krānti-vṛtta* covers it. The axial tilt is what *causes* the ecliptic to be a *circle* on the sky rather than just running along the equator.)

Lesson flow

Warm-up (5 min)

- Prompt: *"Why is it hot in May–June in India and cool in December–January?"*
- Likely wrong answer (very common): *"Because Earth is closer to the Sun in summer and farther in winter."*
- Surprise: Earth is actually slightly *closer* to the Sun in **January** (perihelion ~3 January, 147.1 million km) and slightly *farther* in **July** (aphelion ~4 July, 152.1 million km). India is hot in May–June anyway. So distance is not the answer.
- Real question: *"What else could cause seasons?"*

Core (20 min)

1. **The torch-and-globe demo.** Set up:

    - Place the torch on a desk pointing horizontally.
    - Hold the globe at arm's length, with its axis tilted ~23.5° (use the protractor to check). The N pole should lean either toward or away from the torch.

    Move the globe through four positions, each ~90° around the torch, keeping the axis tilt **fixed in space** (the N pole always points the same direction):

Position	What it represents	Northern Hemisphere	Southern Hemisphere
N pole tilted TOWARD torch	June solstice	Summer (long days)	Winter (short days)
N pole tilted SIDEWAYS (1)	September equinox	Day = night	Day = night
N pole tilted AWAY from torch	December solstice	Winter (short days)	Summer (long days)
N pole tilted SIDEWAYS (2)	March equinox	Day = night	Day = night

2. **What students should see:**
   - When N is tilted toward the torch: the upper half of the globe is brightly lit, and the lit region extends past the N pole. India is well-lit and the Sun is high in the sky. → Summer.
   - When N is tilted away: the upper half is poorly lit, the lit region falls short of the N pole. India is lit at a shallow angle. → Winter.
3. **Two things the tilt does:**
   - **Day length** — when tilted toward, the daytime arc through India is *longer* than the night-time arc. → Longer days in summer.
   - **Angle of sunlight** — sunlight hits a tilted surface at a steeper angle in summer (more concentrated, more heat per square metre) and a shallow angle in winter (more spread out, less heat).
4. **Why does the tilt stay fixed?** Earth's rotation acts like a spinning top — it resists changes in its tilt direction. Over very long times (~25,800 years) the tilt direction wobbles in a slow circle (this is **precession** — Day 8 senior band; we hint at it here only as a footnote).

Activity (15 min) — Build your own tilt model

In pairs, students build a paper "season-checker":

- Take an A4 sheet. Mark a small circle in the centre — that's the Sun.
- Draw Earth's orbit as a big circle around it.
- At four positions on the orbit (top, right, bottom, left), draw a small Earth with its axis tilted to the **right** of vertical (same direction in all four).

- Mark which position is summer in the Northern Hemisphere, which is winter, which are the two equinoxes.
- Label each: June / September / December / March.
- Bonus: mark the Indian subcontinent on each Earth and shade where it's day.

Wrap-up (5 min)

- **Exit ticket on a slip of paper:** "Which is hotter in India in June: the *day length* or the *angle of sunlight* on your skin? Or both? In one sentence."
- Acceptable answer: BOTH — longer days mean more total hours of sunlight, AND the steeper angle means each minute of sunlight delivers more heat per square metre.

Homework

- On the calendar printout, mark the four dates: **~21 March (March equinox), ~21 June (June solstice), ~23 September (September equinox), ~21 December (December solstice).**
- Bring the marked calendar to Day 3.

Student-facing content

> **Earth is tilted.** Its rotation axis is at ~23.5° from straight-up (relative to its orbit around the Sun). And the tilt direction *doesn't change* as Earth orbits.
>
> This single fact causes the seasons:

- When the N pole leans TOWARD the Sun (~June 21) → summer in India.
- When it leans AWAY (~December 21) → winter in India.
- When it leans SIDEWAYS (~March 21 and ~September 23) → day = night everywhere on Earth.

> Surprise fact: Earth is actually *closer* to the Sun in January than in July. Distance is not what causes seasons. Tilt is.

Differentiation

- **One tier up:** What would the seasons be like if Earth's tilt were 0°? (No seasons; equal day/night everywhere all year; equatorial regions hot, polar regions cold permanently.) What if it were 45°? (Extreme seasons — the Arctic Circle would extend down to Delhi's latitude in summer.)
- **One tier down:** Focus on the demo. Just remember: "tilted toward = summer, tilted away = winter, sideways = equinox."

Teacher notes

- The "Earth closer to Sun = summer" misconception is *very* persistent. Highlight that India's June is summer despite Earth being farthest from the Sun in July. The misconception breaks instantly when this is named.
- Some students will ask: *"Then why is India summer in June and not in winter like Australia?"* — Yes, India is in the Northern Hemisphere; Australia is in the Southern.

Same Earth tilt, opposite seasons. (Australia is wearing shorts in December, jumpers in June.)

- The demo will fail visually if the room is too bright. If you can’t darken the room, use a smaller globe and bring the torch closer (3–4 cm from the globe surface).

Citation(s) used in this lesson

- Modern axial-tilt and orbital values from standard NCERT-level astronomy. No primary-source citation needed at this band.